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An Investigation into Accuracy, Computation, and Robustness of Stochastic Linearization of Systems with Saturation Nonlinearities

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ABSTRACT

Quasilinear Control (QLC) is a theory with a set of tools that can be used for the analysis and design of stochastic feedback control systems with nonlinear actuators and sensors. It is based on Stochastic Linearization (SL), a method of linearization that approximates a nonlinearity with an equivalent gain and a bias using statistical properties of its input. While the QLC literature has studied many problems in controls (e.g., performance analysis and controller design), important issues such as accuracy, robustness, and computation of SL have not been fully investigated. This paper fills this gap, with a special focus on feedback systems with saturating actuators, which are commonly encountered in feedback control systems and the QLC literature. Specifically, the accuracy of SL in approximating the saturated response is investigated. A mathematical relationship is established between the saturation authority and the worst-case open-loop accuracy, and the implications on closed-loop accuracy are described. To study the robustness of SL in practical situations, the sensitivities of the equivalent SL gain and bias to system parameters are investigated. Finally, several root-finding algorithms are compared in terms of their computational speed in solving the SL equations, and a novel change of coordinates is presented to improve the performance. The results provide valuable insights into and methods for addressing the computational aspects of SL that are critical for proper design and analysis of nonlinear stochastic feedback control systems.

1. Introduction

1.1. Background and related literature

A major challenge in modern feedback control theory is to effectively handle nonlinear systems affected by stochastic inputs. A typical example of nonlinearity in a feedback system is actuator saturation, which can be caused by, for instance, limits on the flow of fluid through a valve [1]. While there exist theories to analyze and design control systems in their original nonlinear form [2], the more popular approach is to linearize the nonlinear system at an operating point such that the tried-and-tested linear systems theory [3–5] can be applied. However, this method of Jacobian linearization only *locally* approximates the system response and fails to capture it in the presence of stochastic inputs that may drive the system away from the operating point.

An alternative way to linearize a nonlinearity is Stochastic Linearization (SL), which utilizes the probability distribution of the input of the nonlinearity to approximate it by an affine function consisting of an equivalent gain and a bias. This results in a linear

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approximation that, unlike the traditional Jacobian linearization, accurately approximates the first and second moments of the output of the nonlinearity, rendering SL to be a *global* approximation method in a sense. The caveat is that, in a feedback setting, the SL gain and bias depend not only on the nonlinearity itself but also on all exogenous signals and other system parameters. The procedure of SL of a feedback system generally requires solving a system of transcendental equations, which are usually only numerically solvable. SL is similar in principle to the "describing function" methodology [6], as it involves approximating the static nonlinearity by a linear function. However, unlike describing functions, which are intended for the analysis of systems driven by sinusoidal inputs, SL is intended for the analysis of systems driven by stochastic inputs.

There is a significant body of literature in the field of SL. An overview of the prominent works can be found in [7–9]. SL of systems with multivariate nonlinearities is described in [10,11]. The method of "equivalent linearization", which slightly differs from SL in the sense of Kazakov [10] (as applied in this paper) in evaluating the statistical averages, but often used interchangeably with SL in the literature [7,12], was proposed by Caughey in [13]. In [14], the existence and uniqueness of solutions generated by equivalent linearization are discussed. In [15,16], the methods of harmonic balance and SL are combined to approximate nonlinear stochastic systems with pseudo-harmonic behavior. SL for multivariate nonlinearities has been used to analyze and design generic control systems in [17–19].

The recently developed theory of Quasilinear Control (QLC) applies SL to nonlinear *feedback* systems that are frequently encountered by control engineers and extends traditional control techniques like the root-locus design to stochastic systems [20–23]. In [20], the nonlinearities considered are single-variable, symmetric (which, in QLC literature, refers to odd), and appearing only in the actuators and sensors of the control system. In [21,22], QLC was extended to asymmetric (which refers to non-odd in the context of QLC) nonlinearities, but still single-variable and appearing in actuators and sensors. In [24], SL was applied to feedback systems with multivariate nonlinearities appearing anywhere in the loop, not just in the actuators and sensors. While SL has been successfully applied in many examples in the references mentioned above, there are still many open questions surrounding the accuracy, robustness, and computational aspects of it. This paper addresses three of these open questions, as described next. All these are important from the perspective of an SL-based controller design for nonlinear stochastic feedback systems.

1.2. Problems addressed and original contributions

The first problem addressed is related to the *accuracy* of SL in approximating the statistics of the signals in nonlinear feedback systems. Specifically, we assert that the inaccuracies (i.e., approximation errors) of SL in feedback settings can be attributed to two root causes: the inherent inaccuracy of the approximation itself (i.e., open-loop approximation error), and the inaccuracy due to the assumption of Gaussianity in closed-loop settings (i.e., closed-loop error). While there have been studies (e.g., [7,25,26]) on closed-loop accuracy of SL for generic systems, such breakup of the approximation errors was not attempted in the past. In this paper, we delineate this breakup, provide a tight upper bound on the open-loop approximation error, and discuss the implications of this upper bound on the closed-loop accuracy. This information on the accuracy of SL is valuable from a practical perspective, as the error can then be preemptively accounted for during an SL-based design of controllers for the original nonlinear system.

The second problem addressed is the *computation* of SL of a feedback control system. As mentioned above, since SL involves solving implicit equations that generally contain transcendental functions (e.g., the error function), their solution can be arrived at only numerically. Often in practice, numerically solving this system of SL equations is only an intermediate step in solving a larger problem. An example is optimal controller design using SL for a system with saturation (e.g. [1,27]). In such an optimization problem, the SL equations need to be solved at every iteration of the solver. Hence, to ensure fast solve times, it is important to employ numerical algorithms that solve the equations as quickly as possible and with low computational overhead. This has further impacts on stability/performance if the optimization needs to be performed in real-time and updates of controller gains in regular intervals are necessary to maintain adequate performance. In this paper, this issue has been investigated by comparing, through Monte Carlo simulations, six numerical algorithms (specifically, the Bisection, Newton–Raphson, Broyden, Trust-Region Reflective, Trust Region Dogleg, and Fixed Point Iteration methods) for the number of function evaluations required to converge to a solution of the SL equations. It has been found that, in the case of symmetric saturation, the SL functions for most systems are, in fact, contraction maps, but not in the asymmetric case. The best solver in terms of minimizing the amount of computation for solving the SL equations on average in both the symmetric and asymmetric cases is found to be the Trust Region Reflective method. A novel coordinate transformation is also proposed that can improve the success rate of most of the above-mentioned algorithms.

The third problem addressed is related to the *robustness* of SL to system parameters. As alluded to above, the SL gain and bias depend on the statistical properties of the exogenous signals and all other system parameters (including the plant or process being controlled, the controller, and the nonlinear element). If the sensitivity of the SL gain and bias to system parameters is high, a small uncertainty in these parameters may render the linearization (and consequently the designed controller, for example) inaccurate. Hence, it is crucial to quantify which system parameter affects the gain/bias more than others. The sensitivity of SL has been studied in the literature, e.g., in [28]. However, [28] does not compare the relative sensitivities of the SL coefficients to system parameters in a feedback control system and their effect on signal statistics, which could be employed, for example, in optimal controller design. In this paper, we perform the above investigation and show that the SL gain and bias are more affected by those parameters that directly affect the actuator input or its saturation authorities. Also, the sensitivities are found to be higher for systems with an asymmetric saturation nonlinearity than for those with symmetric saturation, and the sensitivity of the SL bias is found to be higher than that of the SL gain. However, the signal statistics are usually not affected much by the SL coefficients, although in the case of asymmetric saturation, it is possible to have a higher sensitivity of the mean of the actuator input and the tracking error.

In sum, this paper investigates the above-mentioned issues of accuracy, computation, and robustness of SL in a typical feedback system with a saturating actuator (a system that is commonly analyzed in various existing QLC literature [23,24,26]). Thus, the original contributions of this paper are as follows:

- A novel metric for the relative accuracy of SL is proposed for the open-loop accuracy of SL, along with upper bounds for the open-loop and, consequently, the closed-loop accuracy of SL.
- A comparison of the computational costs of several common numerical algorithms in solving the SL equations is provided.
- A coordinate transformation is proposed to improve the success rate of most of the above-mentioned algorithms.
- A numerical investigation is carried out into the sensitivity of SL coefficients to system parameters.

1.3. Paper organization

The paper is organized as follows. Section 2 provides a brief review of SL with a focus on saturation nonlinearity. Section 3 discusses open-loop accuracy of SL and its effect on closed-loop accuracy. In Section 4, the computational costs of different numerical algorithms for computing the SL coefficients for systems with symmetric actuator saturation are compared. In Section 5, a sensitivity analysis is performed on the SL coefficients, again for systems with symmetric saturation. Section 6 extends the above investigation to systems with asymmetric saturation nonlinearities. Finally, Section 7 concludes the paper.

2. A brief review of stochastic linearization

In this section, a brief review of SL is provided, and the feedback control system on which the rest of the paper will be based is described. Please refer to [20,21] for details.

2.1. Open-loop

We first consider a nonlinearity in an open-loop setting (i.e., not part of a larger system). Let $u(t)$ be a stationary Gaussian white noise process, with mean μ_u and standard deviation σ_u, and let $f(u)$ be an odd function (i.e., the nonlinearity) with the following properties [11]:

1. $\frac{\partial f}{\partial u}$ exists and is continuous almost everywhere;
2. $|f(u)| < A\exp(u^a)$, $a < 2$, for some $A \in \mathbb{R}$ and any $u \in \mathbb{R}$.

The above assumptions essentially ensure that the function $f'(\cdot)$ is integrable and that the identity, $E[f(u)u] = E[u^2]E[f'(u)]$, which is required for deriving the SL gains, is true. For details, please refer to [11,29].

The goal of SL is to approximate $f(u)$ by $Nu_0(t)+M$, where N and M are constant linearization coefficients, called the *quasilinear gain* and *quasilinear bias* respectively, and $u_0(t) = u(t) - \mu_u$ is the zero-mean part of $u(t)$, such that the mean squared error:

$$\varepsilon(N, M) = E[(f(u) - Nu_0 - M)^2] \tag{1}$$

is minimized. It can be shown [21] that the solution of this problem is:

$$N = E\left[f'(u)\right], \quad M = E\left[f(u)\right] \tag{2}$$

where the above expectations are taken with respect to the probability density function (PDF) of the input process $u(t)$. For a stationary Gaussian process (as assumed above), we need only the mean μ_u and standard deviation σ_u of $u(t)$, as these two completely characterize the process.

Now consider the saturation nonlinearity, which is the most widely studied nonlinearity in the QLC literature:

$$f(u) = \mathrm{sat}_\alpha^\beta(u) := \begin{cases} \beta, & u > \beta \\ u, & \alpha \le u \le \beta \\ \alpha, & u < \alpha \end{cases} \tag{3}$$

where $\alpha < 0$ and $\beta > 0$ are constants and indicate the saturation limits. For this function, the quasilinear gain N and bias M are given by [21]:

$$N = \frac{1}{2}\left[\mathrm{erf}\left(\frac{\beta-\mu_u}{\sqrt{2}\sigma_u}\right) - \mathrm{erf}\left(\frac{\alpha-\mu_u}{\sqrt{2}\sigma_u}\right)\right] =: \mathcal{F}_N(\mu_u, \sigma_u) \tag{4}$$

$$\begin{aligned} M = {} & \frac{\alpha+\beta}{2} - \frac{\beta-\mu_u}{2}\mathrm{erf}\left(\frac{\beta-\mu_u}{\sqrt{2}\sigma_u}\right) + \frac{\alpha-\mu_u}{2}\mathrm{erf}\left(\frac{\alpha-\mu_u}{\sqrt{2}\sigma_u}\right) - \\ & \frac{\sigma_u}{\sqrt{2\pi}}\left[\exp\left(-\left(\frac{\beta-\mu_u}{\sqrt{2}\sigma_u}\right)^2\right) - \exp\left(-\left(\frac{\alpha-\mu_u}{\sqrt{2}\sigma_u}\right)^2\right)\right] =: \mathcal{F}_M(\mu_u, \sigma_u) \end{aligned} \tag{5}$$

where erf(·) is the error function:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

and functions $\mathcal{F}_N(\cdot,\cdot)$ and $\mathcal{F}_M(\cdot,\cdot)$ are introduced to designate the dependence of N and M on μ_u and σ_u.

Quantitatively, the quasilinear gain N from (4) equals the probability that the output, $\mathrm{sat}_\alpha^\beta(u)$, has not been saturated at either its lower limit α or upper limit β [24]. Hence, $0 \le N \le 1$. When the saturation is symmetric, i.e., $\beta = -\alpha$, and u is zero-mean, i.e., $\mu_u = 0$, (4) and (5) reduce to:

$$N = \mathrm{erf}\left(\frac{\beta}{\sqrt{2}\sigma_u}\right), M = 0 \tag{6}$$

2.2. Closed-loop

Consider now the saturation nonlinearity as part of a standard feedback control system depicted in Fig. 1a (the nonlinearity is highlighted in red — typically, it is the actuator). It consists of a linear time-invariant (LTI) plant or process, denoted by $P(s)$, whose output $y(t)$ is to be controlled. The term $P(s)$ refers to the transfer function of the plant, i.e., the ratio of the Laplace transform of its output to that of its input with zero initial conditions, where s is the Laplace transform variable. The plant output, $y(t)$, is measured and compared with a reference, $r(t)$, and the error, $e(t) = r(t) - y(t)$, is input to an LTI controller with transfer function $C(s)$, which computes the control action, $u(t)$, to be input to the saturated actuator. The actuator output, $v(t) = \mathrm{sat}_\alpha^\beta(u(t))$, is then input to the plant $P(s)$.

Since SL requires that all exogenous signals be stochastic, we model the reference $r(t)$ as a stationary colored Gaussian process with mean μ_r and standard deviation σ_r. It is generated as follows: first, a standard white Gaussian noise $w_r(t)$ is passed through a coloring filter, with transfer function $F_{\Omega_r}(s)$, such that $\left\|F_{\Omega_r}(s)\right\|_2 = 1$, where $\|\cdot\|_2$ denotes the $\mathcal{H}_2$-norm, defined for a generic transfer function $G(s)$ as follows:

$$\|G(s)\|_2 := \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} |G(j\omega)|^2 d\omega}$$

and $j = \sqrt{-1}$ is the imaginary unit. Second, the output of the filter is scaled by σ_r and a bias μ_r is added to it to generate $r(t)$.

To find the SL of the nonlinearity $f(u) = \mathrm{sat}_\alpha^\beta(u)$ in such a closed-loop system, using the process described in Section 2.1, we need the PDF of the input, $u(t)$, to the nonlinearity (as indicated in (2)). However, since the closed-loop system is nonlinear, $u(t)$ is not Gaussian, and its PDF is not readily available. Hence, it is typically assumed that the system has been stochastically linearized to that in Fig. 1b (where $m := M - N\mu_{\hat{u}}$ is defined for convenient block diagram manipulation), and the moments arising in the calculation of the linearization coefficients from (4) and (5) (namely, μ_u and σ_u) are approximated by the corresponding moments of the linearized system (namely, the mean $\mu_{\hat{u}}$ and standard deviation $\sigma_{\hat{u}}$ of $\hat{u}(t)$) in Fig. 1b (this procedure is similar to the one used in [7]) for computing N and M. This typically leads to an approximation error, which is small, especially for systems with plants of a low-pass filtering nature [21]. Section 3.3 describes and quantifies this error in detail. Then, through feedback, it can be shown that [21]:

$$\sigma_{\hat{u}} = \left\|\frac{F_r(s)C(s)}{1 + NP(s)C(s)}\right\|_2 \sigma_r \tag{7}$$

$$M = \frac{1}{P_0}\mu_r - \frac{1}{C_0 P_0}\mu_{\hat{u}} \tag{8}$$

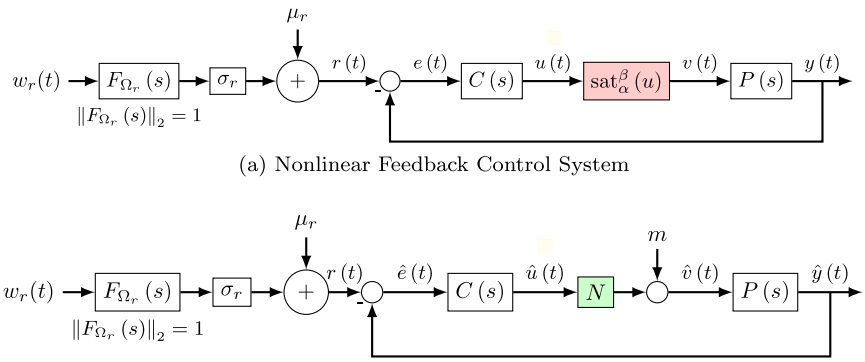
where P_0 and C_0 are the DC gains of $P(s)$ and $C(s)$ respectively (the DC gain of a system with transfer function $G(s)$ is defined as $G_0 = \lim_{s\to 0} G(s)$). Substituting the right hand side (RHS) of (7) for σ_u and the RHS of (8) for M in (4) and (5), and replacing μ_u with $\mu_{\hat{u}}$, we obtain the following two equations in two unknowns, N and $\mu_{\hat{u}}$:

$$N = \mathcal{F}_N\left(\mu_{\hat{u}}, \left\|\frac{F_r(s)C(s)}{1 + NP(s)C(s)}\right\|_2 \sigma_r\right) \tag{9}$$

$$\frac{1}{P_0}\mu_r - \frac{1}{C_0 P_0}\mu_{\hat{u}} = \mathcal{F}_M\left(\mu_{\hat{u}}, \left\|\frac{F_r(s)C(s)}{1 + NP(s)C(s)}\right\|_2 \sigma_r\right) \tag{10}$$

where $\mathcal{F}_N(\cdot,\cdot)$ and $\mathcal{F}_M(\cdot,\cdot)$ are as in (4)–(5). After solving (9) and (10), M can be found by using (8) and m from $m = M - N\mu_{\hat{u}}$. This completes the process of SL of the nonlinear system in Fig. 1a to that in Fig. 1b.

It can be seen that (9) and (10) comprise a system of two transcendental equations in two unknowns. Thus, these equations are only numerically solvable. In this paper, we are thus concerned with investigating the numerical solution of these equations and their properties. Specifically, we have considered two types of nonlinear feedback systems (consistent with the literature of QLC [21,23]): (1) systems having symmetric saturation nonlinearity (i.e., $\alpha = -\beta$) and zero-mean reference input ($\mu_r = 0$), and (2) systems with asymmetric saturation $\alpha \neq -\beta$ and non-zero mean reference input ($\mu_r \neq 0$). For convenience, systems of the first type are henceforth referred to as "symmetric", and those of the second type are referred to as "asymmetric".



(a) Nonlinear Feedback Control System

(b) SL of the System in Fig. 1a

Fig. 1. SL of a feedback system.

3. Analysis of accuracy of stochastic linearization

We begin by investigating the accuracy of SL in this section. Since SL approximates the nonlinear system in a mean-squared sense, there is inevitably an error in the response of the stochastically linearized system compared to that of the original nonlinear system. Consistent with the minimization problem in Eq. (1), this error can be quantified as the mean-squared difference between the stationary response of the original nonlinear system and that of its SL. There are mainly two sources that contribute to the error: (1) the SL of the nonlinearity itself (as in (1), referred to as the open-loop accuracy herein), and (2) the assumption that the input to the nonlinearity in the closed-loop setting is Gaussian and equal to $\hat{u}$ (in Fig. 1b) instead of u (in Fig. 1a). Inspired by Eq. (1), in this section, we introduce a new metric to quantify the open-loop relative accuracy of SL for a saturation nonlinearity. We term this metric the "relative approximation error" and provide its relationship to the actuator authority and the statistics of the input to the nonlinearity. We determine the upper bound on this relative error and show its effect on bounding the closed-loop accuracy. This information on the accuracy of SL is valuable to have from a practical perspective, as the error can then be properly accounted for during an SL-based design of controllers for the original nonlinear system.

3.1. Symmetric case

First, we consider the case of symmetric systems. Recall that the objective of open-loop SL is to minimize the functional (1). Let $f(u) = \mathrm{sat}_{-\beta}^{\beta}(u)$, which is the standard symmetric saturation function with upper limit $\beta > 0$ and lower limit $-\beta$, and $u(t) = u_0(t)$ be a zero-mean stationary Gaussian input. Then, substituting for $f(u)$ and $u_0(t)$ as above and N from (6) in (1), the following expression for the mean squared error, ε, can be obtained in terms of β and σ_u (recall that $M = 0$ in the symmetric case):

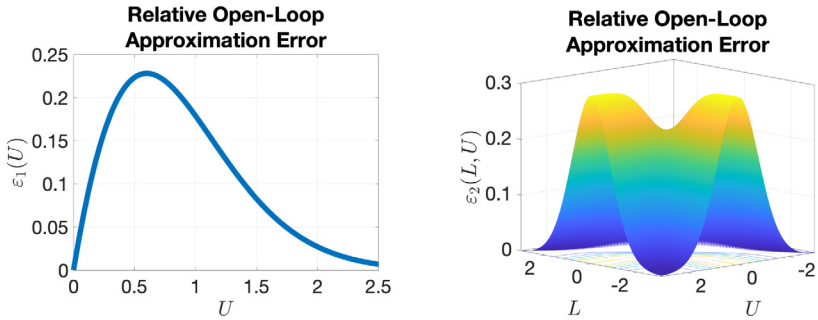
$$\varepsilon(N,0) = -\beta^2 \mathrm{erf}\left(\frac{\beta}{\sqrt{2}\sigma_u}\right) + \beta^2 - \frac{\sqrt{2}\beta\sigma_u e^{-\frac{\beta^2}{2\sigma_u^2}}}{\sqrt{\pi}} - \sigma_u^2 \mathrm{erf}^2\left(\frac{\beta}{\sqrt{2}\sigma_u}\right) + \sigma_u^2 \mathrm{erf}\left(\frac{\beta}{\sqrt{2}\sigma_u}\right) \tag{11}$$

It can be observed from (11) that if we normalize the above by the variance of the input, σ_u^2, the term $\frac{\beta}{\sqrt{2}\sigma_u}$ can be collected and the expression simplified. In addition, this normalization provides a sense of scale of the accuracy relative to the input variance. Thus, letting $U = \frac{\beta}{\sqrt{2}\sigma_u}$, which we term the "saturating factor", we can define the "relative approximation error", ε_1, to be:

$$\varepsilon_1(U) := \sqrt{\frac{\varepsilon(N,0)}{\sigma_u^2}} = \sqrt{2U^2 - \frac{2Ue^{-U^2}}{\sqrt{\pi}} + \left(1 - 2U^2\right)\mathrm{erf}(U) - \mathrm{erf}^2(U)} \tag{12}$$

Fig. 2a shows the variation of $\varepsilon_1(U)$ with U. It can be seen that ε_1 has a maximum value and is thus bounded above. To find the maximum, we evaluate the derivative $\varepsilon_1(U)$ and set it equal to zero. The maximum is found to occur around $U = 0.600$ and the maximum value is around 0.228. The sensitivity of this relative approximation error to changes in system parameters in the feedback case is remarked upon in Section 5.2.

The bell-shaped nature of Fig. 2a can be explained as follows. When $U \gg 0.6$, or the standard deviation of input is very small *compared* to the saturation authority, i.e., $\sigma_u \ll 1.1785\beta$, the error is low (i.e., accuracy is high). This is expected, since in such a case, the output of the saturation function remains practically unsaturated, i.e., $\mathrm{sat}_{-\beta}^{\beta}(u) \to u$. Also, $N = \mathrm{erf}\left(\frac{\beta}{\sqrt{2}\sigma_u}\right) \to 1$ as $\sigma_u \to 0$.



(a) Symmetric Case (b) Asymmetric Case

Fig. 2. Relative open-loop accuracy of SL.

Thus, for large U, it can be mathematically shown that the relative error converges to:

$$\varepsilon_1(U) \to \sqrt{\lim_{\sigma_u \to 0} E\left[\frac{\left(\mathrm{sat}^{\beta}_{-\beta}(u) - Nu\right)^2}{\sigma_u^2}\right]} = \lim_{\sigma_u \to 0} \sqrt{E\left[\frac{(u - Nu)^2}{\sigma_u^2}\right]}$$
$$= \sqrt{\lim_{\sigma_u \to 0} (1-N)^2 \frac{E\left[u^2\right]}{\sigma_u^2}} = \lim_{\sigma_u \to 0} \sqrt{(1-N)^2} = 0$$

where we used the fact that $E\left[u^2\right] = \sigma_u^2$.

As can be seen from Fig. 2a, the relative accuracy is good even when $U \ll 0.6$, i.e., $\sigma_u \gg 1.1785\beta$ (or the standard deviation of input is very large compared to the saturation authority). This is due to the definition of relative accuracy — the division by a large input variance makes the relative accuracy tend toward zero. This makes physical sense as well: the error is small *compared* to the size of the input. In other words, the standard deviation of the actuator input u is much higher than that what is allowed to propagate through the saturation nonlinearity. Since the bounds of the saturation nonlinearity are finite, the error is small *compared* to the size of the input. We now show this mathematically. Consider $\sigma_u \to \infty$. Then $N = \mathrm{erf}(U) \to 0$, and thus:

$$\varepsilon_1(U) \to \lim_{\sigma_u \to \infty} \sqrt{E\left[\frac{\left(\mathrm{sat}^{\beta}_{-\beta}(u) - Nu\right)^2}{\sigma_u^2}\right]}$$
$$= \lim_{\sigma_u \to \infty} \sqrt{E\left[\frac{\left\{\mathrm{sat}^{\beta}_{-\beta}(u)\right\}^2}{\sigma_u^2}\right] + \lim_{\sigma_u \to \infty} E\left[\frac{N^2 u^2}{\sigma_u^2}\right] - 2 \lim_{\sigma_u \to \infty} E\left[\frac{\mathrm{sat}^{\beta}_{-\beta}(u)\, Nu}{\sigma_u^2}\right]} = 0$$

In the last line, the first limit is zero because $\{\mathrm{sat}^{\beta}_{-\beta}(u)\}^2$ is bounded but σ_u^2 is unbounded. The second limit is zero because $E[u^2] = \sigma_u^2$ and $N \to 0$. Finally, the third limit is zero because $E[\mathrm{sat}^{\beta}_{-\beta}(u)u] = \mathrm{erf}(\beta/(\sqrt{2}\sigma_u)) \to 0$, as $\sigma_u \to \infty$.

Thus, in the symmetric case, the open-loop relative error of SL for a saturation function is found to be always better than about 0.228, or 22.8%. Thus, the accuracy is high relative to the size of the input. As shown later, this has consequences in the closed-loop accuracy of SL as well, as it provides an upper bound on how inaccurate the SL approximation can be in a feedback system.

3.2. Asymmetric case

For the case of asymmetric systems, the accuracy is obtained by evaluating the expectation in (1), with $f(u)$ substituted from (3), u_0 as $u - \mu_u$, and N and M substituted from (4) and (5) respectively. Similar to the symmetric case, the expression for accuracy is simplified if we normalize it by the variance of the input u, and define:

$$L := \frac{\alpha - \mu_u}{\sqrt{2}\sigma_u}, \quad U := \frac{\beta - \mu_u}{\sqrt{2}\sigma_u} \tag{13}$$

where L can be termed as the "lower saturation factor" and U the "upper saturation factor" respectively, such that the "relative approximation error" is given by:

$$\varepsilon_2(L,U) := \sqrt{\varepsilon(N,M)/\sigma_u^2}$$
$$= \Big(-\frac{L^2\,\mathrm{erf}^2(L)}{2} + \frac{L^2}{2} + LU\,\mathrm{erf}(L)\,\mathrm{erf}(U) - LU\,\mathrm{erf}(L) + LU\,\mathrm{erf}(U) - LU$$

$$+\frac{Le^{-U^2}\operatorname{erf}(L)}{\sqrt{\pi}}+\frac{Le^{-U^2}}{\sqrt{\pi}}-\frac{Le^{-L^2}\operatorname{erf}(L)}{\sqrt{\pi}}-\frac{U^2\operatorname{erf}^2(U)}{2}+\frac{U^2}{2}-\frac{Ue^{-U^2}\operatorname{erf}(U)}{\sqrt{\pi}}$$
$$+\frac{Ue^{-L^2}\operatorname{erf}(U)}{\sqrt{\pi}}-\frac{Ue^{-L^2}}{\sqrt{\pi}}+\frac{e^{-L^2-U^2}}{\pi}-\frac{\operatorname{erf}^2(L)}{4}$$
$$+\frac{\operatorname{erf}(L)\operatorname{erf}(U)}{2}-\frac{\operatorname{erf}(L)}{2}-\frac{\operatorname{erf}^2(U)}{4}+\frac{\operatorname{erf}(U)}{2}-\frac{e^{-2U^2}}{2\pi}-\frac{e^{-2L^2}}{2\pi}\Bigg)^{\frac{1}{2}} \tag{14}$$

Fig. 2b shows a plot of the above expression for permissible values of L and U, i.e., $L < U$. It can be seen that similar to the symmetric case, the relative open-loop accuracy is always better than a certain level, which in this case, is observed to be the supremum of (14). It can be observed from Fig. 2b that this supremum can be approached asymptotically along either $L = 0$ or $U = 0$. Specifically, it can be found to be:

$$\lim_{U\to\infty}\varepsilon_2(0,U)=\lim_{L\to-\infty}\varepsilon_2(L,0)=\sqrt{\frac{1}{4}-\frac{1}{2\pi}}$$

which has a numerical value of around 0.301, or 30.1%. The sensitivity of this relative approximation error to changes in system parameters in the feedback case is remarked upon in Section 6.4.

The observation can be explained as follows: when $\mu_u \approx \alpha$ or $\mu_u \approx \beta$, the input to the saturation is situated, on average, near the boundary of the regions of the saturation function (which are: $u > \beta$, $\alpha \le u \le \beta$, and $u < \alpha$), and thus the output is more likely to switch between two linear regions, making the resulting linear approximation less accurate. On the other hand, when $\alpha \ll \mu_u \ll \beta$, $\mu_u \ll \alpha$, or $\mu_u \gg \beta$, the input is situated, on average, in a linear region of the function, and hence, the resulting linear approximation is more accurate. The good accuracy for large σ_u, when L and U both approach zero, is due to division by σ_u^2, which makes the relative accuracy smaller (similar to the symmetric case).

Thus, the relative accuracy of open-loop SL for an asymmetric saturation is also bounded above and always better than 30.1%, which is higher than the 22.8% in the symmetric case. Moreover, as the magnitude of L or U increases, which can be brought about by a decrease in σ_u, the accuracy improves. As shown next, this has consequences in the closed-loop system, as it bounds the closed-loop accuracy of SL as well.

3.3. Implications for closed-loop accuracy

Consider the closed-loop system shown in Fig. 1a and its SL in Fig. 1b. Analogous to the previous subsections, we can define the closed-loop "relative approximation error" of SL as the square root of the mean squared error between the output of the nonlinearity in Fig. 1a and its SL in Fig. 1b, normalized by the actuator input variance:

$$\varepsilon_3 := \sqrt{\frac{E\left[\left(\operatorname{sat}_\alpha^\beta(u)-N\hat{u}_0-M\right)^2\right]}{\sigma_{\hat{u}}^2}} \tag{15}$$

We choose to normalize by the variance of the actuator input $\hat{u}(t)$ from the stochastically linearized system as opposed to the nonlinear system since its value is readily available in terms of system parameters, whereas the PDF of the $u(t)$ in the nonlinear system is not readily available. We now proceed to show that this closed-loop relative approximation error is bounded above, leveraging the results from the previous subsections. First, through the algebraic manipulations shown below, it can be observed that the closed-loop RMSE satisfies:

$$\begin{aligned}&\sqrt{E\left[\left(\operatorname{sat}_\alpha^\beta(u)-N\hat{u}_0-M\right)^2\right]}\\&=\sqrt{E\left[\left(\left\{\operatorname{sat}_\alpha^\beta(u)-\operatorname{sat}_\alpha^\beta(\hat{u})\right\}+\left\{\operatorname{sat}_\alpha^\beta(\hat{u})-N\hat{u}_0-M\right\}\right)^2\right]}\\&\le\sqrt{E\left[\left\{\operatorname{sat}_\alpha^\beta(u)-\operatorname{sat}_\alpha^\beta(\hat{u})\right\}^2\right]}+\sqrt{E\left[\left\{\operatorname{sat}_\alpha^\beta(\hat{u})-N\hat{u}_0-M\right\}^2\right]}\end{aligned} \tag{16}$$

by Minkowski inequality [30]. The last line of (16) is profound, as it prominently delineates the breakup of the closed-loop approximation error into two components: the first represents the error due to the Gaussianity assumption on the input to the nonlinearity in the closed-loop setting (i.e., $\hat{u}$ in Fig. 1b is equal to u in Fig. 1a), while the second represents the open-loop approximation error due to the SL of the nonlinearity itself. We now further simplify the right hand side as follows. Since $\alpha \le \operatorname{sat}_\alpha^\beta(u) \le \beta$ and $\alpha \le \operatorname{sat}_\alpha^\beta(\hat{u}) \le \beta$, $\left|\operatorname{sat}_\alpha^\beta(u)-\operatorname{sat}_\alpha^\beta(\hat{u})\right| \le \beta-\alpha$. Hence,

$$\sqrt{E\left[\left\{\operatorname{sat}_\alpha^\beta(u)-\operatorname{sat}_\alpha^\beta(\hat{u})\right\}^2\right]}\le\beta-\alpha \tag{17}$$

Dividing (16) by $\sigma_{\hat{u}}$ and using (13) and (17), we obtain:

$$\varepsilon_3 \le \sqrt{2}(U-L)+\varepsilon_2 \tag{18}$$

Thus, the closed-loop relative approximation error ε_3 is found to be bounded above by the sum of the open-loop relative approximation error, ε_2 (in the symmetric case, $\varepsilon_2 = \varepsilon_1$), and a term that depends on the difference of the saturation factors. Since ε_3 depends directly on ε_2, for the same value of $\sigma_{\hat{u}}$, a smaller value of ε_2 leads to a smaller upper bound for ε_3. In other words, if the open-loop relative accuracy is good, so will the closed-loop relative accuracy. The open-loop relative approximation error ε_2 has already been shown to be bounded above in previous subsections, while the term $\sqrt{2}(U-L)$ provides a bound on the closed-loop relative approximation error due to the Gaussianity assumption on u. Moreover, as $\sigma_{\hat{u}}$ becomes larger, or the saturation authorities become more restrictive, the upper bound of the relative approximation error in closed-loop approaches that in the open-loop case.

In summary, the analyses in the preceding subsections show that SL of a saturation nonlinearity can produce relative mean squared errors less than 30.1% in the actuator output but relative accuracy improves with a smaller variance in the input. Eqs. (12) and (14) provide a way to analytically determine the statistics of the actuator input that leads to the largest open-loop mean squared error for a given actuator authority. Finally, it is shown that open-loop relative accuracy being bounded above also results in the closed-loop accuracy being bounded above by a value that is dependent on the saturation authority and the actuator input. This information can be used to design controllers that take into account the possible error in the closed-loop case, for example, involving a robust optimization problem whose solution is robust to linearization error. This is a topic of future research.

4. Computation of stochastic linearization

We investigate the computation of SL coefficients in this section. As mentioned in Section 2.1, the SL Eqs. (9) and (10) are transcendental and, hence, only numerically solvable. For practical considerations, it is important to employ numerical solvers that can solve them as quickly and efficiently as possible. In this section, some common numerical algorithms are compared for the number of function evaluations until convergence. Since analytic solutions are not possible, we rely on Monte Carlo simulations (similar to [21,23]) to carry out this investigation.

Only symmetric systems are considered in this section since asymmetric systems give rise to unique computational challenges and results. They are analyzed separately in Section 6. Specifically,

- The case of asymmetric saturation requires solving a set of *two* simultaneous equations, and is thereby more complicated, whereas the case of symmetric saturation requires solving only *one* equation.
- When the equation-solving (i.e., root finding) algorithms are applied to the asymmetric case (Section 6.2), most of the systems require many "extra tries" (see below), indicating that this case is not numerically robust. This prompts the coordinate transformation in Section 6.3 so that the number of extra tries required is lowered.

4.1. Algorithms examined

The numerical algorithms that are considered, both in this section and the next for the asymmetric case, are briefly described below:

Bisection method: The bisection method [31,32] finds a root of an equation (or a system of equations) by reducing the search interval in half in an iterative manner, based on the function values at the endpoints of the interval. Since the quasilinear gain $N \in [0,1]$, this algorithm naturally has a bounded initial search interval and can be applied to find SL of symmetric systems. However, the bisection algorithm cannot be applied to the case of asymmetric systems because $\mu_{\hat{u}}$ does not belong to a compact set, which means that the initial search space is unknown. The QLC literature has historically been using this method to solve SL equations [20].

Newton–Raphson method: The Newton–Raphson method is an iterative algorithm that uses the gradient of the function to choose a step direction for the next iteration, with the weight of this direction decided by the functional value at that point [33]. This algorithm works for any sized system of equations, making it applicable for both symmetric and asymmetric cases. The downsides are that the gradient must be available for computation at every iteration and that convergence is guaranteed only when the initial guess is sufficiently close to the solution.

Broyden's method: This is a quasi-Newton method that uses the same procedure as the Newton–Raphson method, but with the gradient being approximated by the slope of the line segment between the current and previous guesses [34]. Therefore, it typically has a lower computational overhead as compared with Newton–Raphson, since it does not need to evaluate the gradient explicitly, nor must it evaluate the system of functions more than once at each iteration.

Trust-region reflective method: This method is similar to the Newton–Raphson Method but is more robust in situations where the gradient is singular, or the initial guess is far from the solution. It involves solving a quadratic program around the current best solution (i.e, around a *trust region*) to determine the next step. It is used for solving an unconstrained optimization problem [35]. This algorithm is used by default in MATLAB's `lsqnonlin`. Recall that a root-finding problem $g(x) = 0$ for a vector function $g : \mathbb{R}^n \to \mathbb{R}^n$ can be posed equivalently as an unconstrained optimization problem of minimizing $\sum_{i=1}^{n} g_i(x)^2$, where $g_i(x)$ is the ith component of $g(x)$ [36]. Hence, this algorithm is applied by reformulating the SL root-finding problem as an unconstrained minimization problem and solving the resulting optimization problem using this algorithm. Note that other trust-region techniques can be used as well. For example, the Trust Region Dogleg is a similar algorithm used for solving equations [37]. However, since in our studies, the results obtained with the Trust Region Dogleg algorithm were similar to that obtained using the Trust Region Reflective method, we have omitted the former from our paper.

Fixed Point Iteration: This method leverages the fact that for a contractive function with a fixed point, repeated function evaluations can be used to converge to the fixed point [38]. Specifically, given the fixed point problem $x = g(x)$, where g a contraction, the following iterative procedure can be used to find the solution: $x_{n+1} = g(x_n)$ starting from any initial guess.

Table 1
Parameter ranges used for random system generation.

Parameter	Value range
Controller gain, K_c	$(0, 50)$
Plant gain, K_p	$(0, 5)$
Saturation boundary, β	$(0, 10)$
1st order controller time constant, τ_c	$(0.01, 100)$
1st order plant time constant, τ_p	$(0.01, 100)$
2nd order plant natural frequency, ω_p	$(0.01, 100)$
2nd order plant damping factor, ζ_p	$(0, 2)$
Reference signal standard deviation, σ_r	$(0, 5)$

4.2. Problem setup

Systems of the form shown in Fig. 1a were chosen for the Monte Carlo study, with only the symmetric cases being considered (for now). A random controller $C(s)$ and plant $P(s)$ were generated for each system, along with random saturation authorities (keeping $\alpha = -\beta$) and reference standard deviation σ_r (with $\mu_r = 0$). To incorporate a variety of situations, three combinations of plant and controller classes were considered: (1) a constant controller with a first-order plant (19), (2) a first-order controller with a first-order plant (20), and (3) a constant controller with a second-order plant (21).

$$C(s) = K_C, \quad P(s) = \frac{K_P}{\tau_P s + 1} \tag{19}$$

$$C(s) = \frac{K_C}{\tau_C s + 1}, \quad P(s) = \frac{K_P}{\tau_P s + 1} \tag{20}$$

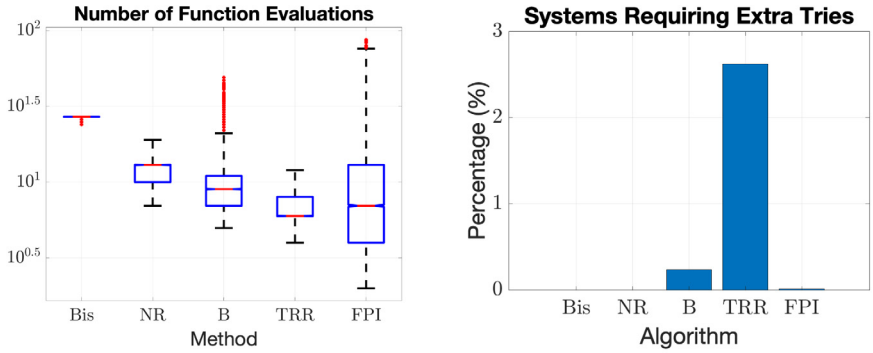
$$C(s) = K_C, \quad P(s) = \frac{K_P \omega_p^2}{s^2 + 2\zeta_p \omega_p s + \omega_p^2} \tag{21}$$

Table 1 lists the sampling ranges for all these parameters. They were chosen such that the distribution of the quasilinear gain N was more-or-less equally skewed toward both $N = 0$ and $N = 1$, fairly representing systems ranging from both highly nonlinear to fully linear systems. For example, it was observed through a Monte Carlo simulation that values of $\beta = -\alpha < 5$ result in N being more closer to 0 and $\beta = -\alpha > 15$ result in N being more closer to 1. All parameters were sampled uniformly, except for the first-order time constants and the second-order natural frequency, which were sampled logarithmically. To obtain data from a large number of systems, each class of system was sampled 10,000 times; thus a total of 30,000 systems were considered. For each system, (9) was solved for the unknown, N (note that (10) is not required to be solved since the saturation is assumed to be symmetric), with a relative convergence tolerance of 1×10^{-8} on the unknown variable, i.e., $(|N_2 - N_1|)/(1 + |N_1|) < 1 \times 10^{-8}$, where N_2 is the value of N at the current iteration and N_1 that at the previous iteration. Systems with phase margin (which is a metric used to quantify the stability and robustness of a control system), PM $< 40°$ were rejected, as they are not indicative of a well-designed control system (see Remark 1 in the next subsection). For algorithms that require the gradient (i.e., the Newton–Raphson and Trust-Region Reflective methods), the gradient was provided by finite differencing. This was done by small perturbations, unlike the finite differencing performed in the Broyden method. For all the algorithms, the numbers of function evaluations (including Jacobian evaluations) were collected. The number of iterations was capped to 400, and if an algorithm was unable to find a solution with a given initial condition, it was allowed to retry with a different random initial condition. The number of such "retries" was capped to 100; if an algorithm failed even after 100 tries, that system was considered a "failure" for the algorithm (since the algorithm would effectively be unusable for practical purposes if it requires many number of tries).

4.3. Results

The Monte Carlo experiment revealed that the SL equation for symmetric feedback systems is, in fact, a contraction map (for all randomly generated systems). This is an important and profound discovery into the nature of the SL equations and paves the way for future research into the topological and algebraic properties of this equation. Also, it highlights that the fixed-point iteration can be successfully employed to solve the SL equation.

The results from the Monte-Carlo simulation are illustrated in Figs. 3a and 3b. Fig. 3a shows a box plot of the number of function evaluations for different algorithms and Fig. 3b shows the number of systems that required extra tries to converge to a solution. All algorithms eventually found the solution, i.e., there were no "failures". From 3(b), it can be seen that less than 3% of the systems required extra tries. In the x-axis of Fig. 3a, "Bis" stands for bisection, "NR" for Newton–Raphson, "TRR" for Trust-Region Reflective, and "FPI" for Fixed Point Iteration. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to the most extreme data points not considered outliers, and the outliers are plotted individually using the red '°' symbol. Any data point that is greater than $x + 1.5y$ or less than $x - 1.5y$ is considered an outlier, where x is the median and y the inter-quartile range. From Fig. 3a, it can be seen that the bisection algorithm (which is commonly used in the QLC literature) performs significantly worse than all of the other algorithms on average, with mean iterations being more than two times greater than those of others, although it has the smallest spread. On average, the Trust-Region Reflective method requires the least number of function evaluations, followed closely by Fixed Point Iteration. However, the Fixed



(a) Box Plot of Function Evaluations (b) Systems requiring extra tries

Fig. 3. Computation of SL.

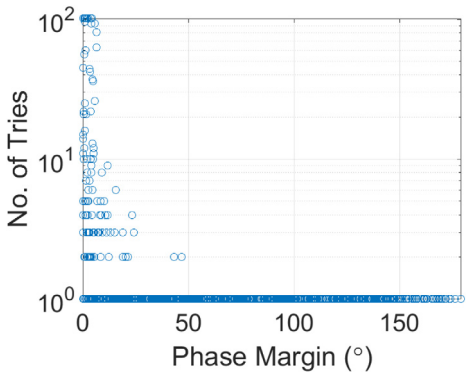


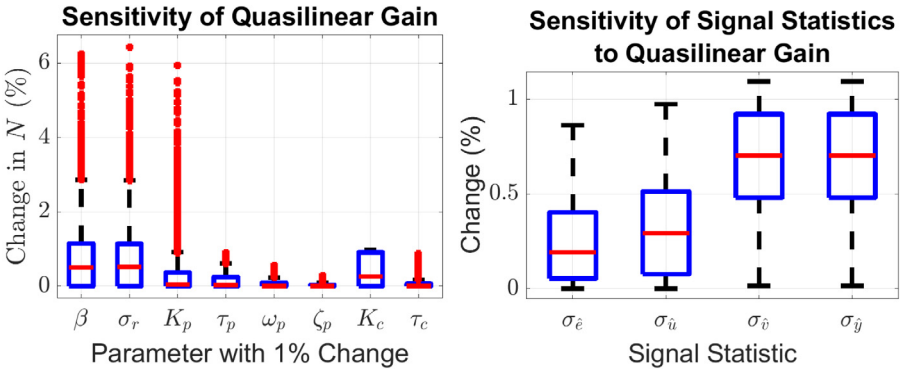
Fig. 4. Number of tries for fixed point iteration method.

Point Iteration method has a bigger spread in the number of evaluations. This is because the gradients of the functions are close to one, although they are always contractions. From Fig. 3b, it can be seen that the Trust Region Reflective method requires more tries in some cases, whereas the Newton–Raphson method does not, although both have almost the same spread in the number of function evaluations. We conclude that, if the goal is to ensure certainty of a solution, the Newton–Raphson method is the best, and if the goal is to ensure computational speed most of the time, the Trust Region Reflective is the best.

Remark 1. In a separate study, we considered systems with PM $< 40°$. We found that, while none of the algorithms failed (i.e., required tries more than 100) for PM $> 40°$, in the case of PM $< 40°$, a few (about 0.3%) of the systems simulated failed to converge for Fixed Point Iteration, indicating that they may not be contraction maps. The number of tries for some systems increased for PM $< 40°$, as shown in Fig. 4, and more tries were required for those systems which have smaller phase margin.

5. Robustness of SL

Recall that the quasilinear gain N and bias M found by solving (9) and (10) depend on system components like the plant $P(s)$ and the controller $C(s)$, as well as on exogenous signal statistics like the mean and standard deviation of the reference, $r(t)$. This dependence on various parameters may render these SL coefficients sensitive to these parameters, which may be problematic. For example, in optimal control settings (e.g., the linear quadratic regulator or LQR [39]), the variance of the actuator input, σ_u^2 (or $\sigma_{\hat{u}}^2$ in the SL reformulation), is often penalized in the objective function of the optimal control problem. If the SL coefficients are highly sensitive to system parameters, it may render the optimal control problem numerically sensitive (or even inaccurate) to these parameters as well. Thus, the knowledge of these sensitivities can help to understand which parameters of the system must be estimated/measured more accurately for reliable computation of the quasilinear gain/bias and, hence, the optimal controller. To obtain sensitivity data for a large range of feedback systems, a Monte Carlo investigation was performed. This was done since an analytic investigation of comparative sensitivities is not possible due to the highly transcendental and nonlinear nature of the SL equations. Similar to the previous section, again, only symmetric systems are studied here, and asymmetric systems are separately investigated in Section 6. This is because in the case of robustness for asymmetric systems, the sensitivities to *both* the quasilinear gain N and bias term, m, need to be considered rather than only N in the symmetric case. Also, since in the asymmetric case, the exogenous signals have a non-zero mean, the sensitivities of the *means* of the signals must be considered apart from only the *standard*



(a) Sensitivity of Quasilinear Gain to System Parameters

(b) Sensitivity of Signal Statistics to Quasilinear Gain

Fig. 5. Sensitivity of SL coefficients.

deviations, as considered in this section. Finally, it is found in Section 6 that the sensitivities in the case of asymmetric saturation are, in general, larger.

5.1. Problem setup

The same systems as in Section 4.2 were considered here for the Monte Carlo study. A 1% change in various system parameters was applied (representing, for example, typical numerical tolerance in the measurement or estimation of the parameters), the quasilinear gain N was re-calculated, and the change from the previous value recorded. Further, to investigate the effect of a change of N on the computation of the statistical properties of the various signals, the standard deviations of the tracking error, $\sigma_{\hat{e}}$, the actuator input, $\sigma_{\hat{u}}$, the actuator output, $\sigma_{\hat{v}}$, and the plant output, $\sigma_{\hat{y}}$, were recorded, both with the original value of N at the solution of the SL equations and with a 1% increase in that value of N.

5.2. Results

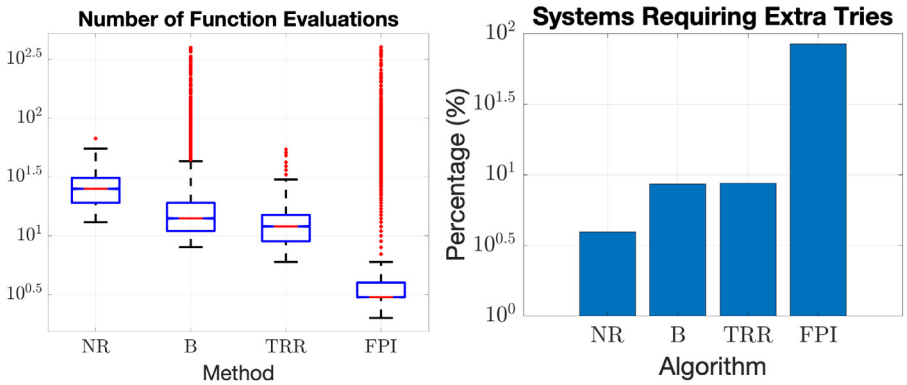
Fig. 5a shows a box plot of the percentage change in the solved quasilinear gain N, i.e., $|\Delta N/N| \times 100\%$, where ΔN refers to the change in N after *re-solving* the SL equations, for a 1% change in various system parameters, i.e., $|\Delta\theta/\theta| \times 100\% = 1\%$, where $\Delta\theta$ is a change in the system parameter θ.

We have removed cases where $N < 0.1$ since the percentage change metric (i.e., $|\Delta N/N| \times 100\%$) does not convey useful information when the original value N is zero or close to zero (also, we are interested only in the *comparative* sensitivities of N to various parameters). From Fig. 5a, it can be seen that the overall change in N is less than 1% on average for all parameters (note that this is less than the 1% change introduced in the parameters). Also, while there is not much effect of the plant or controller dynamics (i.e., the time constant, damping factor, or natural frequency), there is a considerable effect of the reference standard deviation σ_r, actuator authority β, and the controller and plant gain on the quasilinear gain. This observation can be explained by the fact that these parameters directly affect the magnitude of the input to the saturation, or the bounds of saturation. The probability of saturation is therefore directly influenced by these parameters. Moreover, since β, σ_r, K_c, and τ_c are usually known, it is the plant gain K_p that needs to be accurately measured/estimated. Knowing these sensitivities, one essentially also knows the level of potential variability in the quasilinear gain due to uncertainties in the model parameters. This can further be leveraged in, for example, an optimization problem to design robust controllers, which is a topic for future work.

Fig. 5b shows the effect of a 1% change in N at the solution (i.e., $|\Delta N/N| \times 100 = 1\%$, where ΔN is a change *introduced* in the quasilinear gain N at the solution) on standard deviations of various signals (again, removing instances where the original value of the standard deviation is less than 0.1). That is, the y-axis of Fig. 5b shows $|\Delta\sigma_q/\sigma_q| \times 100\%$, where $\Delta\sigma_q$ refers to the change in the standard deviation of the particular signal q, σ_q, due to a 1% change in N. It can be observed that the standard deviations of the signals are affected by less than 1% on average (note that this is less than the 1% change introduced in the parameters). This indicates that even if the quasilinear gain is sensitive to system parameters, this will generally contribute even lesser to a change in the signal statistics. This is a useful property to have for an optimization problem that may involve such statistics for designing optimal controller gains.

Remark 2. The trends for sensitivity for different parameters do not change for systems with PM $< 40°$, although it was noticed that for some systems the sensitivities were higher than in the case with PM$> 40°$.

Remark 3. The sensitivity of the open-loop approximation error (12) for the stochastically linearized system was found to be, in general, close to zero for all system parameters, except for the plant time constants, in which case, the sensitivity was higher.



(a) Box Plot of Function Evaluations (b) Systems requiring extra tries

Fig. 6. Computation of SL coefficients.

6. The case of asymmetric saturation

In this section, the investigation of robustness and computational efficiency of SL is continued for nonlinear stochastic systems with asymmetric saturation and a nonzero mean reference signal.

6.1. Problem setup

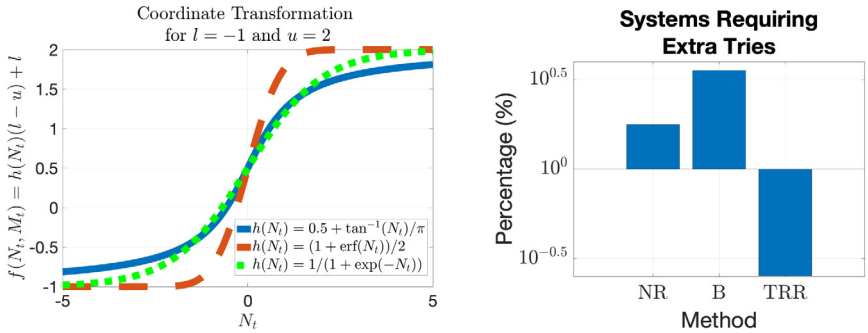
For investigating the computation and robustness of SL in the following subsections, Monte Carlo studies were conducted on the same classes of systems that were generated in Section 4.2, but with $\alpha \neq \beta$, α being generated uniformly from $(-10, 0)$. The reference mean μ_r was sampled uniformly from $(-5, 5)$. Both (9) and (10) were now solved for N and $\mu_{\hat{u}}$. For assessing computational performance, all algorithms mentioned in Section 4.1 were considered except the Bisection algorithm (as explained in Section 4.1). Moreover, in the case of Fixed Point Iteration, (10) was not used in the form shown since it is not a fixed point equation. Instead, $\mu_{\hat{u}}$ was replaced by $C_0(\mu_r - P_0 M)$ and $\sigma_{\hat{u}}$ by RHS of (7) in (9) and (10), and the resulting equations were instead solved for the fixed point (N, M). Systems in which $L > 0$ or $U < 0$ (where L and U are defined in (13)) were removed since, for a well-designed control system, it is expected that the nonlinear actuator will not be affected by signals whose values are outside the saturation bounds on *average*. Also, as before, systems with a phase margin of $< 40^{\circ}$ were rejected, as they are not typically employed in practical settings.

6.2. Results for computation of stochastic linearization

Fig. 6a shows the number of function evaluations required for convergence for different algorithms and Fig. 6b the number of systems that required extra tries, including failed systems (i.e., no. of tries > 400). It can be seen that the number of systems requiring extra tries significantly increased from the symmetric case (Fig. 3b). This is expected since the asymmetric case has a more complicated system of transcendental equations. Although from Fig. 6a, the number of function evaluations is found to be the least on average in the Fixed Point Iteration method (for systems where the method converged successfully), more than 95% of the systems failed (i.e., required tries more than 400) using this method, indicating that the RHS of (4) and (5) are not contractions for most systems. This is in contrast to the symmetric case, where the SL equation is a contraction for all systems. The Newton–Raphson method requires a comparatively lesser number of tries but the most number of function evaluations. Upon analyzing the cause of a large number of tries or failures in the Newton–Raphson method, it was found that in some iterations, the quasilinear gain was negative (as previously shown, N must satisfy $0 \leq N \leq 1$). Negative values of N are non-physical and are an artifact of the numerical solver. To remedy this, a coordinate transformation is proposed to prevent the solver from choosing non-physical values, as mentioned in the next subsection.

6.3. Improving computation through coordinate transformation

The idea of coordinate transformation is to allow solvers to have access to a larger space for variables than what is allowed in the base coordinate system, such that function evaluations do not result in undefined values. To ensure that the quasilinear gain N does not become negative in (9) and (10), $(N, \mu_{\hat{u}})$ can be transformed to a new coordinate system (N_t, M_t), which is defined through the following mapping: $N = f(N_t, M_t)$ and $\mu_{\hat{u}} = g(N_t, M_t)$, where f and g are transformations that satisfy these conditions: (1) f and g are bijective and differentiable such that the Jacobian of $\left[f\left(N_t, M_t\right) \quad g\left(N_t, M_t\right)\right]^{\mathrm{T}}$ is invertible, (2) N_t and M_t can take all real values, and (3) The range of f is the interval $[0, 1]$, which is the range of allowable values for N in the case of SL for the saturation function. The first condition ensures that there is a one–one mapping between the coordinate systems and the function



(a) Graphs of common $f(N_t)$ (b) Systems requiring extra tries

Fig. 7. Coordinate transformation for SL.

does not become flat to avoid numerical issues. The second condition ensures that the solvers have access to an infinite space for the variables so that functional values do not become undefined.

Several forms of $f(N_t, M_t)$ are possible, including $f(N_t, M_t) = h(N_t)$, where $h(N_t)$ is one of the following (note that the function h is introduced, in addition to f, to simplify the discussion below, as we will show): $h(N_t) = 0.5 + \arctan(N_t)/\pi$, $h(N_t) = (1 + \mathrm{erf}(N_t))/2$, and $h(N_t) = 1/(1 + e^{-N_t})$. However, as $N_t \to \pm\infty$, these functions $h(N_t)$ tend to 0 or 1 asymptotically, which may result in numerical issues if $N \approx 0$ or $N \approx 1$ as the function becomes flat at these extremes. Hence, the third requirement on f is modified to: the range of f is $[0 - \epsilon, 1 + \epsilon] \cap \mathcal{N}_s$, where $\epsilon \geq 0$ is a small number and $\mathcal{N}_s$ is the set of values of N that makes the closed-loop SL system stable (it can be verified that $\mathcal{N}_s = \{N : N > -1/P_0C_0\}$ for the classes of systems considered in Section 4.2, where P_0 and C_0 are the DC gains of the plant and controller respectively). For $\epsilon \neq 0$, this is essentially done by defining f such that $f(N_t, M_t) := h(N_t)(l - u) + l$, where $h(N_t)$ are as mentioned above, l represents the lower limit of $\mathcal{N}_s$, $\mathcal{N}_{s,\min}$, and $u = 1 + \epsilon$. See Fig. 7a for graphs of these modified functions. For $g(N_t, M_t)$, possible functions are $g(N_t, M_t) = C_0(\mu_r - M_tP_0)$ (where M_t is actually M, as mentioned in Section 2.2), $g(N_t, M_t) = (C_0\mu_r - C_0M_tP_0)/(1 + C_0f(N_t)P_0)$ (where M_t is actually m) or $g(N_t, M_t) = M_t$ (when M_t is actually $\mu_{\hat{u}}$ and is not transformed).

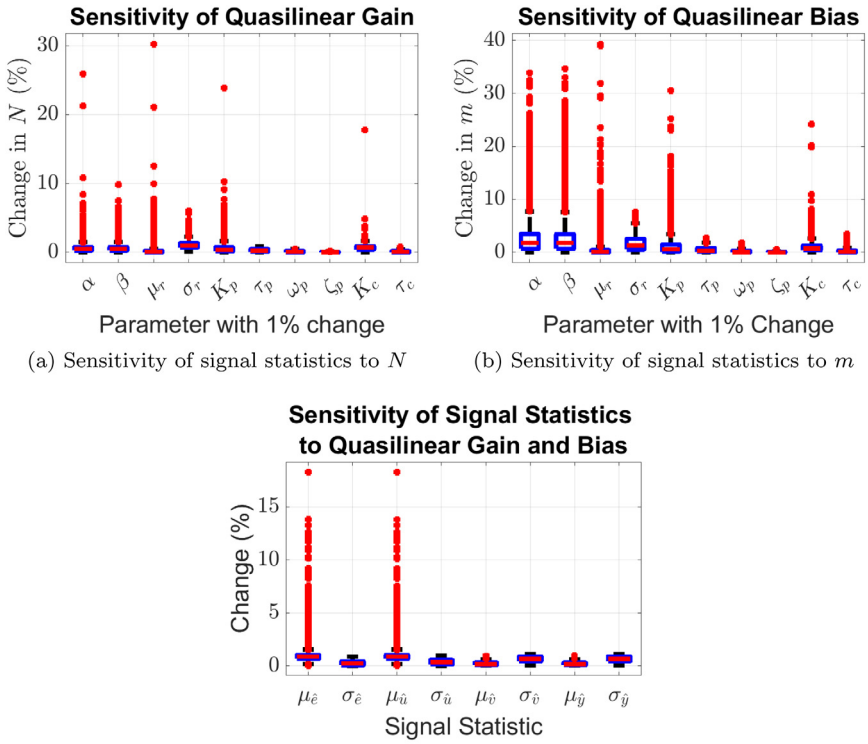
In our investigation on the same systems considered in Section 6.2, it was found that out of all the above-mentioned combinations of the forms of $f(N_t, M_t)$ and $g(N_t, M_t)$, the forms of $f(N_t, M_t) = (0.5 + \tan^{-1}(N_t)/\pi)(l - u) + l$ and $g(N_t, M_t) = M_t$, with some $\epsilon > 0$, leads to the best results. The Monte Carlo experiment mentioned in Section 6.2 was performed again with these transformed coordinates for $\epsilon = 1$. The fixed point iteration method was not attempted again since the coordinate transformations destroy the contraction property. This is because the derivative of the inverse of arctan, erf and exp functions are greater than unity for the values of quasilinear gain considered. The results are shown in Fig. 7(b). It can be seen that the number of systems requiring extra tries has reduced significantly for all methods (compared to Fig. 6b), with the lowest for the Trust-Region Reflective method. The number of function evaluations was similar on average.

Hence, for SL of asymmetric systems, it is our recommendation to use the transformations $f(N_t, M_t) = (0.5 + \tan^{-1}(N_t)/\pi)(l - u) + l$ and $g(N_t, M_t) = M_t$, with an $\epsilon \approx 1$, $u = 1 + \epsilon$, l as defined above, and the optimization-based Trust-Region Reflective method for quick convergence.

6.4. Results for robustness analysis

Recall that the procedure of SL in the case of asymmetric systems results in a value of N and $\mu_{\hat{u}}$ from (9) and (10). From the value of $\mu_{\hat{u}}$ found, both M and $m = M - N\mu_{\hat{u}}$ can be found (in fact, from any of $\mu_{\hat{u}}$, M and m, the other two can be found). Physically, $\mu_{\hat{u}}$ refers to the mean of the actuator input, M the mean of the actuator output, and m an external bias term that needs to be injected into the system as shown in Fig. 1b. Since out of $\mu_{\hat{u}}$, M, and m, only m represents an external input, its value can be manipulated physically in a practical system, and hence, we performed the sensitivity analysis concerning N and m. The results for the sensitivity of the solved quasilinear gain N and $m = M - N\mu_{\hat{u}}$ to a 1% change in various system parameters (the definition of % change is the same as in Section 5.2) are shown in Figs. 8a and 8b respectively. The sensitivities of signal statistics (μ denoting the mean and σ the standard deviation of the signal) to a 1% change introduced in N and m at the solution are shown in Fig. 8c. As before, those cases are removed where the magnitude of the parameters whose sensitivities are being found are less than 0.1 since the percentage change metric does not reliably portray the change when the original value is close to zero. From Fig. 8a, it can be seen that compared to the symmetric case, the sensitivity has increased for many systems to more than 20%. Also, from both Figs. 8a and 8b it can be observed that while the sensitivities of N and m to system parameters are, on average, close to zero, there are many systems where the bias term m is highly sensitive to the system parameters. Again, as in the symmetric case, it is found that sensitivities are higher for those parameters that directly affect the magnitude of the actuator input and its saturation authority, e.g., the controller or plant gains, and the actuator authorities.

From Fig. 8c, it can be seen that while the sensitivities of the signal statistics to the computed N and m are, on average, low, although the sensitivity of the mean of the tracking error, $\hat{e}$, and of the actuator input, $\hat{u}$, is higher for many systems. This is due to the



(a) Sensitivity of signal statistics to N

(b) Sensitivity of signal statistics to m

(c) Sensitivity of signal statistics to N and m.

Fig. 8. Sensitivity of SL coefficients.

effect of feedback, and the fact that the magnitude of the mean of the error is typically small for a reference tracking problem. The results indicate that if an optimization problem uses the mean of the tracking error or the actuator input in the objective function, then the numerical tolerance of the solver should be carefully chosen and the parameters of the system accurately identified to avoid sensitivity issues. Note that such issues do not arise if only the standard deviation of the error and that of the actuator input are used in the objective function, as done, for example, in a standard LQR problem. We have not found a clear correlation of the outliers with known system parameters, and that is a topic for future investigation.

Remark 4. Similar to the symmetric case, the sensitivity of the open-loop approximation error (14) for the stochastically linearized system was found to be, in general, close to zero for all system parameters, except for the plant time constants, in which case, the sensitivity was higher.

7. Conclusions and future work

In this paper, the open-loop accuracy, robustness, and computational performance of SL for a typical stochastic nonlinear feedback system with both asymmetric and symmetric saturation have been investigated. From the study on open-loop accuracy, a mathematical relationship has been established between the actuator authority and the worst-case error, and its implications on closed-loop accuracy described. From the study on the robustness of SL to system parameters, it is found out that the SL coefficients are more sensitive in the case of asymmetric saturation than in the case of symmetric saturation, and that the sensitivity is more toward those parameters that affect the saturation directly (e.g., gain, actuator authority, signal statistics). Also, in the case of asymmetric saturation, the mean of the actuator input and tracking error is more sensitive to changes in SL coefficients. Finally, from the study on computational efficiency, it is found that the Broyden method is best suited for applying SL to symmetric saturation in a reference tracking problem since it requires a low computational effort. In the case of asymmetric saturation, it is found that a coordinate transformation can lead to significant improvements in the number of tries required by an algorithm, and the Trust Region Reflective method requires the least number of tries. For both asymmetric/symmetric saturation, it is found out that fixed point iteration is not an effective method for solving the SL gain/bias.

Future work on this topic includes comparing other quasi-Newton methods (since other algorithms in the quasi-Newton class maybe even more effective than Broyden's Method, but were not examined during this study), studying robustness and accuracy for multivariate nonlinear stochastic systems, incorporating sensitivity analysis/results in a robust optimal control problem, investigating the cause of outliers found in the sensitivity analyses of our paper, SL of systems with nonlinearities other than

saturation, and closed-loop accuracy for systems with low phase margin. In the broader context of QLC, we also intend to investigate further topics such as SL for discrete-time systems and the effect of SL on controllability and observability of systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Brahma S, Almassalkhi MR, Ossareh HR. A stochastic linearization approach to optimal primary control of power systems with generator saturation. In: 2018 IEEE Conference on Control Technology and Applications (CCTA). 2018, p. 982–7. http://dx.doi.org/10.1109/ccta.2018.8511401.

[2] Isidori A. Nonlinear Control Systems. Third Edn.. Springer; 1995.

[3] Wonham W. On a matrix riccati equation of stochastic control. SIAM J Control 1968;6(4):681–97. http://dx.doi.org/10.1137/0306044.

[4] Krener AJ, Isidori A. Linearization by output injection and nonlinear observers. Systems Control Lett 1983;3(1):47–52. http://dx.doi.org/10.1016/0167-6911(83)90037-3, URL: http://www.sciencedirect.com/science/article/pii/0167691183900373.

[5] Hauser J, Sastry S, Kokotovic P. Nonlinear control via approximate input-output linearization: the ball and beam example. IEEE Trans Automat Control 1992;37(3):392–8. http://dx.doi.org/10.1109/9.119645.

[6] Khalil HK. Nonlinear Systems. third. Prentice Hall; 2002.

[7] Socha L. Linearization Methods for Stochastic Dynamic Systems, Vol. 730. Springer Science & Business Media; 2007, http://dx.doi.org/10.1007/978-3-540-72997-6.

[8] Roberts JB, Spanos PD. Random Vibration and Statistical Linearization. Dover Publications; 2003.

[9] Elishakoff I, Crandall SH. Sixty years of stochastic linearization technique. Meccanica 2017;52(1–2):299–305. http://dx.doi.org/10.1007/s11012-016-0399-x.

[10] Kazakov I. Generalization of statistical linearization method to multidimensional systems. Avtomat Telemekh 1965;26(7):1210–5.

[11] Atalik TS, Utku S. Stochastic linearization of multi-degree-of-freedom non-linear systems. Earthquake Eng Struct Dyn 1976;4(4):411–20. http://dx.doi.org/10.1002/eqe.4290040408.

[12] Socha L, Pawleta M. Are statistical linearization and standard equivalent linearization the same methods in the analysis of stochastic dynamic systems?. J Sound Vib 2001;248:387–94. http://dx.doi.org/10.1006/jsvi.2001.3705.

[13] Caughey TK. Equivalent linearization techniques. J Acoust Soc Am 1963;35(11):1706–11.

[14] Spanos PD, Iwan W. On the existence and uniqueness of solutions generated by equivalent linearization. Int J Non-linear Mech 1978;13(2):71–8.

[15] Di Matteo A, Spanos P, Pirrotta A. Approximate survival probability determination of hysteretic systems with fractional derivative elements. Probab Eng Mech 2018;54:138–46.

[16] Spanos P, Zhang Y, Kong F. Formulation of statistical linearization for MDOF systems subject to combined periodic and stochastic excitations. J Appl Mech 2019;86(10).

[17] Kazakov I, Malczikow S. Analysis of stochastic systems in state space. Nauka, Moskwa 1975;67:79.

[18] Kazakov I. Statistical theory of control systems in state space (Russian). Moscow, Izdatel'stvo Nauka 1975;(1975):432.

[19] Naumov B. Theory of nonlinear automatic systems: Frequency methods. 1972.

[20] Ching S, Eun Y, Gokcek C, Kabamba PT, Meerkov SM. Quasilinear control: Performance analysis and design of feedback systems with nonlinear sensors and actuators. Quasilinear Control: Performance Analysis and Design of Feedback Systems with Nonlinear Sensors and Actuators. Cambridge University Press; 2010, http://dx.doi.org/10.1017/cbo9780511976476.

[21] Kabamba P, Meerkov S, Ossareh H. Stochastic linearisation approach to performance analysis of feedback systems with asymmetric nonlinear actuators and sensors. Internat J Control 2015;88(1):65–79. http://dx.doi.org/10.1080/00207179.2014.938300.

[22] Guo Y, Kabamba PT, Meerkov SM, Ossareh HR, Tang CY. Quasilinear control of wind farm power output. IEEE Trans Control Syst Technol 2015;23(4):1555–62. http://dx.doi.org/10.1109/tcst.2014.2363431.

[23] Huang W-P, Brahma S, Ossareh HR. Quasilinear Control of Systems with Time-Delays and Nonlinear Actuators and Sensors. In: American Control Conference (ACC), Philadelphia; 2019:.

[24] Brahma S, Ossareh HR. Quasilinear Control of Feedback Systems with Multivariate Nonlinearities. In: 58th IEEE Conference on Decision and Control (CDC), Nice, France; 2019:.

[25] Ching S, Meerkov SM, Runolfsson T. GaussianIzation of random inputs by filtering plants: The case of Poisson white and telegraph processes. In: Decision and Control (CDC), 2010 49th IEEE Conference on. Ieee; 2010, p. 2650–5.

[26] Gokcek C, Kabamba PT, Meerkov SM. Disturbance rejection in control systems with saturating actuators. In: American Control Conference, 2000. Proceedings of the 2000. Nonlinear Anal 2000;40:213–26.

[27] Gokcek C, Kabamba PT, Meerkov SM. An LQR/LQG theory for systems with saturating actuators. IEEE Trans Automat Control 2001;46(10).

[28] Socha L, Soong T. Sensitivity and linearization techniques in analysis of non-linear stochastic systems. J Sound Vib 1992;156(1):79–97.

[29] Roberts JB, Spanos PD. Random Vibrations and Statistical Linearization. 1990.

[30] Minkowski H. Geometrie Der Zahlen. Creative Media Partners, LLC; 2018, URL: https://books.google.com/books?id=KTXrugEACAAJ.

[31] Burden RL, Faires JD. 2.1 the bisection algorithm. In: Numerical Analysis. 3rd edn.. PWS Publishers; 1985.

[32] Wood GR. The bisection method in higher dimensions. Math Program 1992;55(1–3):319–37.

[33] Ypma T. Historical development of the Newton–raphson method. SIAM Rev 1995;37(4):531–51. http://dx.doi.org/10.1137/1037125.

[34] Broyden CG. Quasi-Newton methods and their application to function minimisation. Math Comp 1967;21(99):368–81, URL: http://www.jstor.org/stable/2003239.

[35] Coleman TF, Li Y. An interior trust region approach for nonlinear minimization subject to bounds. SIAM J Optim 1996;6(2):418–45.

[36] Dennis Jr JE, Schnabel RB. Numerical Methods for Unconstrained Optimization and Nonlinear Equations. SIAM; 1996.

[37] Powell MJD. A FORTRAN subroutine for solving systems of nonlinear algebraic equations.. Comput J 1968;24:87–91.

[38] Rhoades B. Some fixed point iteration procedures. Int J Math Math Sci 1991;14(1):1–16.

[39] Kwakernaak H, Sivan R. Linear Optimal Control Systems, Vol. 1. Wiley-interscience New York; 1972.